

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

4 Sebastian Bobadilla-Suarez
 `sebastian.bobadilla.s@gmail.com`

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6 **Abstract**

7 We present a machine-checked synthesis of established physical
8 results into a single conditional route from the finiteness of a local-
9 ized system’s phase space to a unified cognitive fixed point. No new
10 fundamental physics is proposed: Kibble’s classification of topolog-
11 ical defects, Landauer’s erasure bound, the bound of Still and col-
12 leagues relating dissipation to nonpredictive information, the thresh-
13 old $K_c = 2D$ for the noisy mean-field Kuramoto model, and Banach’s
14 fixed-point theorem are all imported. What is new is that they are
15 stated in one formal language, their mutual consistency mechanically
16 checked, and the points at which the route relies on stipulation rather

17 than derivation located exactly. One conditional theorem in Lean 4
 18 composes eight named hypotheses to its endpoint; the unproved steps
 19 are its arguments rather than sentences about it, and they are jointly
 20 satisfiable rather than jointly empty. Two links are marked rather
 21 than argued around: the identification of the posited continuum ker-
 22 nel with the endogenous electromagnetic field of cortex is an empirical
 23 commitment with four stated failure conditions, and the identification
 24 of a global section of the probability sheaf with the unity of experience
 25 is a stipulation. The framework is falsifiable in one place that is its
 26 own. Recovery from sudden awakening should follow a delayed sig-
 27 moid with a square-root cusp rather than an exponential, because the
 28 slow variable gating recovery is astrocytic volume rather than cortical
 29 phase realignment, and throughout that recovery the concentration
 30 and the coherence of the phase distribution must collapse onto the
 31 parameter-free curve $r = I_1(a)/I_0(a)$.

32 **1 Introduction**

33 The pursuit of a physical theory of consciousness typically involves search-
 34 ing for the biological correlates of mind or treating subjective experience
 35 as an abstract computational algorithm. This paper asks a more funda-
 36 mental question: if we assume only basic physical primitives—continuous
 37 symmetries, geometry, and thermodynamics—is the emergence of a unified,
 38 perceiving entity a natural physical attractor?

39 We propose a framework in which the structural properties of conscious-
 40 ness are organized as candidate physical attractor states, running from con-
 41 tinuous symmetries and boundaries, through dissipation and physical reso-
 42 nance, to the macroscopic electromagnetic unity proposed by field theories
 43 of mind. This is not a derivation from basic principles: the formal result is
 44 a conditional composition schema whose eight unproved arrows are explicit
 45 assumptions. To avoid critiques of “math theatre,” we delineate between lit-
 46 eral physical mechanisms (thermodynamic erasure via Landauer’s principle,
 47 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 48 using Lean 4 formalization as our mathematical anchor. We adopt a perspec-
 49 tive aligned with Russellian monism: objective measurements of macroscopic
 50 electromagnetic fields constitute the third-person “outside,” while the sys-
 51 tem’s internal physical deformation to reach structural resonance constitutes
 52 the first-person intrinsic experience.

53 **What is new here, and what is not**

54 A framework paper about consciousness invites a particular suspicion, and it
 55 is better answered at the start than at the end. **No new physics is derived**
 56 **here.** Every physical result the chain invokes is established, and most of
 57 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 58 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 59 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
 60 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative

61 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
 62 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
 63 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
 64 Bell and Crooks relating dissipation to nonpredictive information (Still et
 65 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
 66 dissipated work with the relative entropy between forward and time-reversed
 67 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,
 68 1922). A physicist reading this paper will recognize all of it. Conceding the
 69 point costs nothing.

70 What is new is of a different kind, and it is a smaller claim. These results
 71 are stated in one formal language, their mutual consistency mechanically
 72 checked, and the places where the chain relies on stipulation rather than
 73 derivation located exactly. The development declares **no axioms**: `#print`
 74 `axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. Ev-
 75 ery physical postulate is a field of a class each model discharges for itself,
 76 and every class is inhabited by an explicit witness, so no theorem is vac-
 77 uous. The links form a single conditional composition theorem — **chain**
 78 in `extttChain.lean` — with **eight named hypotheses**: four formalization
 79 gaps, two modelling assumptions, two physical commitments. Its conclu-
 80 sion `extttUnifiedSelf` says the sheaf-glued section is the unique fixed point
 81 of self-prediction, rather than `Unity` and `Self` being unrelated. The theo-
 82 rem `chain_hypotheses_jointly_satisfiable` discharges all eight on one
 83 toy substrate, ruling out inconsistency without evidence that cortex realizes

84 the composition.

85 The schema exposes two missing arrows: the capacity bound implies the
86 heat cost directly, skipping the boundary, and neither of the two theorem-
87 arrows is one the framework’s usual narration draws.

88 A hypothesis of that theorem is worth counting only if it cannot be dis-
89 charged by ignoring it, and two of the eight are about the existence of a
90 structure: a predictive structure at the third arrow, a thermodynamic cover
91 at the seventh. Both structures are inhabited in the development indepen-
92 dently of the premise — a two-bit predictive system, a two-patch cover —
93 so bare existence would be an arrow whose proof discards its own hypoth-
94 esis. Each therefore demands a structure tied to the object the premise is
95 about. The third asks for a predictive structure whose thermal scale is the
96 register’s own temperature, whose dissipated work is the register’s own Lan-
97 dauer heat, and whose wasted memory is not zero; the seventh asks for a
98 cover whose patch coupling is at least the mean-field constant the coherent
99 regime names, and whose equilibrium configuration is the limit of a relax-
100 ation on that coupling rather than a configuration the instance is placed at.
101 Each demand has a regression beside it: a predictive structure that dissipates
102 nothing fails the first (`frozenSystem_not_of_eraser`), and a cover reached
103 by relaxation at unit coupling fails the second at $K = 3$ (`trioCover_not_`
104 `reachedByRelaxation_three`).

105 The framework is not thereby unfalsifiable, and one of its predictions is
106 its own rather than a rival’s. Section 12 argues that recovery from sudden

107 awakening should follow a *delayed sigmoid* rather than an exponential, be-
 108 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
 109 clearance, on a timescale of seconds to minutes, while the cortical rhythms
 110 whose realignment is usually taken to constitute the recovery move in mil-
 111 liseconds. That is a claim about functional form; it is fittable against data
 112 of a kind already collected; and a purely electrical account of the same re-
 113 covery predicts a different form. Because a delayed sigmoid is what any slow
 114 saturating gate would give, that section also states the sharper claim the
 115 same mathematics supports: at every instant of the recovery the concentra-
 116 tion and the coherence of the phase distribution must lie on the fixed curve
 117 $r = I_1(a)/I_0(a)$, the same curve for every subject, with no free parameters
 118 and with the threshold read off the ratio a/r rather than fitted.

119 **2 The chain, and the status of each of its links**

120 The machine-checked dependency accounting is carried out in Lean 4, build-
 121 ing on recent efforts to formalize consciousness and brain semantics such as
 122 the ‘OnlyOne’ system (Scherf, 2026) and sheaf-theoretic models of neural
 123 manifolds (Inoué, 2026). Figure 1 shows the schema with the status of every
 124 node *and of every arrow* marked, and Table 1 states in seven rows what
 125 each link assumes. The complete claim-by-claim map to Lean identifiers is
 126 Table S1 in the supplement; the main text does not depend on reading it.

127 One result about the formalization itself is worth stating here rather than

128 in the supplement, because it is what makes the accounting mean anything.
129 An axiom that is inconsistent silently makes every theorem downstream of
130 it vacuous while leaving the build green, and the failure has one shape: *an*
131 *axiom that constrains a symbol it does not itself bind*. Such a declaration
132 is not a postulate about one system but a constraint on every model of the
133 class the symbol belongs to, including models built to violate it, and a class
134 that has such a model is thereby inconsistent. Three tempting global axiom
135 shapes — pointwise vacuum minimization with the energy functional as an
136 implicit argument, a Landauer heat identity equating a free class field with a
137 bath-entropy difference for every instance, and an entropy-production bound
138 asserted against all pairs of distributions — each prove **False**, from that one
139 root cause. The rule that prevents it is the rule we follow throughout: *a*
140 *physical postulate that mentions a class field must be a field of that class,*
141 *never a standalone axiom quantified over all instances*. A postulate is then
142 an obligation each model discharges, not a global claim about every model.
143 The supplement gives the three refutations and the corresponding witness
144 constructions, which are what keep the classes inhabited and their theorems
145 non-vacuous.

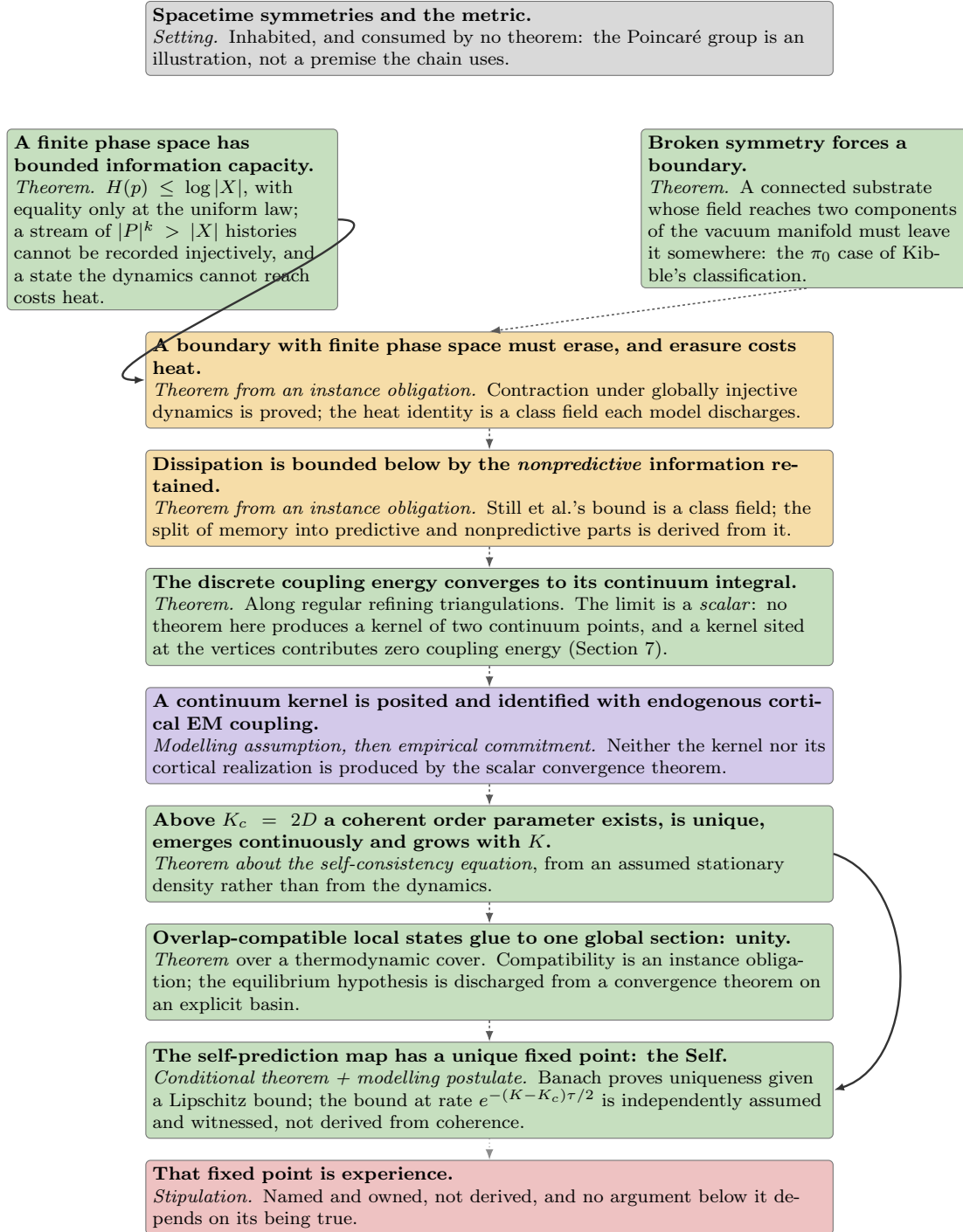


Figure 1: The conditional composition schema, with the status of every node and the distinction between the two kinds of arrow. *Nodes.* ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ setting, consumed by no theorem. *Arrows.* A **solid** arrow is a Lean theorem. A **dashed** arrow is a named hypothesis of `chain` (`extttChain.lean`): an implication between the two node statements which the development does not prove and which is therefore an argument the theorem takes. A **dotted** ar

Claim	Status	What is assumed
A finite phase space has bounded information capacity	Theorem	Finiteness of the state space, and nothing else — no symmetry group, no metric. $H \leq \log X $, with equality only at the uniform law. On an infinite state space the step from “unreachable state” to “erasure” fails, and the failure is exhibited rather than assumed away.
Broken symmetry forces a boundary	Theorem	Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.
Erasure costs heat	Theorem + class field	Global system–bath dynamics injective (Liouville). The heat identity <code>heat_eq</code> is discharged per instance, not asserted for all models.
Dissipation is bounded below by nonpredictive information	Theorem + class field	Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on. On a finite register with a bipartite environment the bound is <i>derived</i> from Landauer’s, given the identification of the wasted memory with the erased entropy — an identification no theorem supplies, and one a reversible register cannot satisfy with any waste at all.
The discrete coupling energy converges to its continuum integral; the field carrying that coupling is the cortical EM field	Theorem; then empirical 9	Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The limit is a scalar: no theorem produces a kernel of two continuum points, and a kernel sited at the triangulation’s vertices contributes zero coupling energy on an atomless substrate. The physical identification is a commitment with stated failure conditions (Section 7).
Above $K_c = 2D$ a unique coherent state exists, and overlap-compatible patches glue into a unique global section	Theorem	Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated. Gluing needs a cover at the potential minimum

146 3 Symmetries as setting, and the capacity of 147 a finite phase space

148 Our starting premise relies on fundamental geometry. Continuous spacetime
149 symmetries act upon field states, and the invariance of the system’s total
150 energy under that action is recorded by the class `SymmetryInvariantAction`
151 (`Phase1Primitives.lean`). Physically, it is such an invariance that Noether’s
152 theorem converts into the conservation of energy and momentum. The scope
153 of the *formal* claim is narrower and we state it plainly: the invariance is a
154 class field, an assumption a model must discharge, and no conserved quantity
155 is ever constructed from it. There is no Noether theorem in our Lean, and
156 nothing downstream depends on one. The class is inhabited by a double-
157 well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime with the $\mathbb{Z}_2$ action
158 $v \mapsto \pm v$, whose vacuum manifold $\{-1, 1\}$ is proved degenerate with the two
159 minima exchanged by the symmetry. That is symmetry breaking exhibited
160 rather than assumed; it is still not a conserved quantity. Symmetries and the
161 metric are therefore *setting* rather than content, and Figure 1 marks them
162 as such.

163 The other half of the premise is not setting. Because phase space is finite,
164 any localized physical system possesses a mathematical limit on the informa-
165 tion it can embody at any moment, and three theorems in `Phase1PhaseSpaceCapacity.lean`
166 carry it, whose only hypothesis about the system is that its state space is
167 finite.

168 The first is the capacity bound. A probability distribution p on a fi-
 169 nite state space X satisfies $H(p) \leq \log |X|$ (`shannon_entropy_le_log_card`),
 170 with equality if and only if p is uniform (`shannon_entropy_eq_log_card_`
 171 `iff`). The “only if” half is the part worth stating: $\log |X|$ is the exact capac-
 172 ity rather than a convenient over-estimate. The proof is Gibbs’ inequality
 173 against the uniform law, with the $p(x) = 0$ case carrying the slack that makes
 174 the equality condition sharp.

175 The second says the limit binds. Let the boundary absorb a stream of
 176 perturbations drawn from an alphabet P ; a stream of length k has $|P|^k$
 177 possible histories, and once $|X| < |P|^k$ the map from histories to final states
 178 cannot be injective (`absorb_not_injective`), so two distinct histories leave
 179 the system in the same state (`source_entropy_exceeds_capacity`). The
 180 update map is arbitrary and may be reversible at every individual step; the
 181 collision is forced by counting alone, which is the sense in which the premise,
 182 and not the model of the dynamics, does the work.

183 The third is what the thermodynamic step consumes. On a finite state
 184 space, a map that fails to reach every state is not injective (`is_erasure_of_`
 185 `not_surjective`) — which is exactly the erasure condition Landauer’s bound
 186 requires — so any update of a finite phase space leaving some state unreach-
 187 able dissipates strictly positive heat (`finite_phase_space_dissipates`). On
 188 an infinite phase space the step simply fails: the successor map on $\mathbb{N}$ misses
 189 0 and is injective. That failure is recorded beside the witnesses, so the finite-
 190 ness hypothesis is fenced rather than assumed to be load-bearing. None of

191 this derives the finiteness of phase space; finiteness is assumed, as in any
192 statistical-mechanical treatment. What is removed is the gap between as-
193 suming the premise and using it.

194 4 Symmetry breaking and the inevitability of 195 boundaries

196 A perfectly symmetric universe is the uniform vacuum; no distinct entities
197 exist. For any dynamic process to occur, the continuous symmetries of the
198 background field must undergo spontaneous symmetry breaking, and topol-
199 ogy then dictates the creation of defects such as domain walls. This neces-
200 sitates a physical boundary—a localized topological distinction between an
201 “inside” and an “outside.” Such a defect is necessary to define a separate
202 entity but not sufficient for agency: the boundary is relational, defined by
203 the twisted geometry of the field surrounding it, and must be coupled with
204 non-equilibrium thermodynamics to exhibit active behaviour.

205 That first claim is about topology, and it needs a different argument from
206 the energy claim beside it. The energy half — an energy-minimizing field sits
207 in the vacuum manifold almost everywhere (`pointwise_vacuum_of_global_`
208 `min`) — together with the observation that a defect *obstructs* extension over
209 a contractible interior, produces no defect: both are statements about a
210 field already given. What produces one is `exists_notMem_of_connected-`
211 `ComponentIn`: if the substrate is connected and a continuous field takes val-

ues in two *different connected components* of the vacuum manifold M , then
somewhere the field is not in M at all. That excluded region is the wall.
This is the π_0 case of Kibble’s classification (Kibble, 1976), and the proof
is elementary in exactly the way it is elementary in the physics literature:
the continuous image of a connected space is connected, and a connected
subset of M lies inside one component of it. No hypothesis mentions energy,
a potential, or a symmetry.

We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are available in Mathlib’s
homotopy library and not taken, because the argument this paper runs uses
domain walls and nothing else. The witness is the double well, where the vac-
uum manifold is *computed* to be exactly $\{-1, 1\}$ rather than merely known to
contain both minima, and its disconnection is proved in the form the theorem
consumes. `wellV_domain_wall` then quantifies over *every* continuous field on
any connected substrate reaching both minima, with no formula for the field
in the statement; that the class is non-empty is checked separately, since a
theorem quantified over all fields is worthless if none satisfies its hypotheses.

5 The thermodynamic constraint

This topological boundary is subjected to gauge perturbations from the “out-
side” environment. Because the internal phase space is finite, it cannot ab-
sorb an unbounded sequence of incoming perturbations and must overwrite

old internal states. By Landauer’s principle (Landauer, 1961), the erasure of physical information irreversibly decreases internal entropy, mandating that the system exhaust heat into the environment.

Norton and Shenker have argued that *logical* erasure (many-to-one on abstract computational states) need not dissipate heat if implemented by an injective physical process (Norton, 2005; Shenker, 2000). Our formalization addresses this by defining erasure as a non-injective mapping on the *physical* state space and formalizing the global system–bath evolution as strictly injective (Liouville). Heat dissipation is then a consequence of local state-space contraction under globally reversible dynamics. The finiteness premise enters exactly here: on a finite phase space, a state becoming unreachable is the same as non-injectivity, and `finite_phase_space_dissipates` is the result. The Norton–Shenker distinction is preserved rather than blurred: an update that keeps the incoming perturbation on record is injective and dissipates nothing, while one that overwrites it is not. Converting the combinatorial statement to heat requires the class field `StatisticalMechanics.heat_eq`, discharged by an explicit one-bit erasure system.

6 Prediction as the thermodynamic cost of memory

A dissipative structure is driven by its environment. Write X_t for the state of the structure and S_t for the environmental signal driving it, and let the

254 signal evolve by a dynamics of its own on which the structure does not act.
 255 Two quantities are then defined at each step: the *memory* $I(X_t; S_t)$, and the
 256 *predictive information* $I(X_t; S_{t+1})$, which is what that same state says about
 257 the signal's next value.

258 **What the entropy-production bound does not give.** There is a nat-
 259 ural route from entropy production to predictive processing, and it does not
 260 work. Modelling the environment as a stochastic driving force $P(\text{ext})$ and
 261 the internal transitions as $Q(\text{int})$, entropy production is bounded below by
 262 $D_{KL}(P||Q)/\Delta t$ — this is Gibbs' inequality, carried in Lean as an instance
 263 obligation. Rearranged it reads $D_{KL} \leq \Delta t \sigma$, so a system descending on σ
 264 lowers a *bound* on the divergence, and it is tempting to conclude that the
 265 internal transition probabilities come to mirror the environment's statistical
 266 symmetries. They do not, because a driven system does not relax to $\sigma = 0$:
 267 it relaxes to a non-equilibrium steady state at strictly positive entropy pro-
 268 duction, and the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number.
 269 No limit may be read off it.

270 **Where predictive processing comes from.** Still, Sivak, Bell and Crooks
 271 (Still et al., 2012) prove that for a system driven by an environmental signal
 272 the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

273 The bracket is the *nonpredictive* information: what the structure retains
 274 about the signal's present that says nothing about its future. Memory
 275 that fails to predict is not merely useless to the structure; it is thermo-
 276 dynamic waste, quantified in energy. The bound is carried as the class field
 277 `PredictiveDissipation.still_bound`, for the same reason as every other
 278 physical postulate here: its `axiom` form, quantified over all joint laws with
 279 the dissipated work fixed, is refutable, and the development refutes it.

280 What is *derived* is everything downstream. First, the bracket is non-
 281 negative: $X_t \rightarrow S_t \rightarrow S_{t+1}$ is a Markov chain, so the data processing inequal-
 282 ity gives $I(X_t; S_{t+1}) \leq I(X_t; S_t)$ (`predictiveInfo_le_mutualInfo`). Second,
 283 dissipation is bounded below by a non-negative quantity, so $W_{\text{diss}} \geq 0$ follows
 284 rather than being assumed (`dissipatedWork_nonneg`). Third, the inference
 285 runs in the valid direction: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$, so any bound
 286 on dissipation is immediately a bound on wasted memory (`nonpredictive_`
 287 `le_dissipation`). In the idealised quasi-static case every retained bit is
 288 predictive.

289 The no-back-action hypothesis is load-bearing and we exhibit what breaks
 290 without it. In the formalization it is not an assumption stated beside the
 291 theorem but the shape of the evolution itself: the signal's step is the parallel
 292 kernel `id || κ`, so no channel from X_t to S_{t+1} exists. A Markov kernel on the
 293 *pair* which copies the system's state into the signal carries an independent
 294 two-bit law to a perfectly correlated one, raising mutual information from
 295 zero to a positive value: an agent that steers its environment can know

296 more about that environment's future than about its present, and Still's
 297 bound is not about such an agent. Both regimes are exhibited over one
 298 correlated two-bit joint law. With the signal frozen, every remembered bit
 299 is predictive and the instance may dissipate nothing, with the memory itself
 300 proved strictly positive; with the signal redrawn from its stationary law at
 301 each step, the entire memory is nonpredictive and the instance is *forced* to
 302 dissipate $k_B T$ times all of it. Kawai, Parrondo and Van den Broeck (Kawai
 303 et al., 2007) supply the corresponding equality — average dissipated work is
 304 exactly $k_B T$ times the relative entropy between forward and time-reversed
 305 phase-space densities — carried here as a predicate on an instance, which
 306 replaces the abstract dissipation field with a quantity an experiment can
 307 estimate (`nonpredictive_le_arrow`).

308 **Landauer's heat is the budget Still's bound is drawn against.** The
 309 two dissipation accounts in the chain — a finite register charged for de-
 310 stroying information, a driven structure charged for keeping information it
 311 cannot use — are accounts of one physical quantity, and the bridge module
 312 `Phase3_LandauerBridge.lean` identifies them. `PredictiveDissipation.`
 313 `ofLandauer` takes a register with a bipartite environment and returns a
 314 predictive structure whose thermal scale is that register's temperature and
 315 whose dissipated work *is* its Landauer heat, with `still_bound` derived from
 316 `landauer_bound` rather than assumed a second time. What the construction
 317 asks for is one inequality, and it is the physics: the memory the structure

318 wastes is no larger than the entropy the update erases, $(I_{\text{mem}} - I_{\text{pred}}) \leq$
 319 $S(\text{id}) - S(t)$. That the wasted bits and the cleared bits are the same bits is
 320 an identification no theorem supplies, and it is not free: on a *reversible* up-
 321 date the erased entropy is zero, so the inequality forces the wasted memory
 322 to vanish outright (`nonpredictive_eq_zero_of_injective`). A register that
 323 keeps a record rather than clearing one cannot be given a budget for waste —
 324 which is the Norton–Shenker distinction of the previous section, arriving from
 325 the informational side. On the one-bit eraser the two quantities are equal:
 326 the register destroys $\log 2$ of entropy, the correlated two-bit law wastes $\log 2$
 327 of memory, and the bound is met with equality (`landauerSystem_tight`).
 328 The measure-theoretic statement of the same principle is checked against the
 329 combinatorial one rather than left beside it: at counting measure on a finite
 330 register the volume-contraction bound is exactly $T(S(\text{id}) - S(t))$ (`landauer_`
 331 `heat_bound_count_eq`).

332 **What this does not establish.** It does not deliver the Free Energy Prin-
 333 ciple. Still’s bound says that prediction is thermodynamically favoured; it
 334 says nothing about hierarchical generative models, variational free energy,
 335 or Bayesian inference. The Free Energy Principle (Friston, 2010; Friston &
 336 Kiebel, 2009; Hohwy, 2013; Clark, 2016; Spisák & Friston, 2026) and the
 337 General Resonance Theory of Consciousness (Hunt & Schooler, 2019) are
 338 consonant with what is proved here, which is a weaker relation than deriva-
 339 tion and the only one we assert. Nor does it solve the symbol grounding

340 problem (Harnad, 1990): maximal predictive information is compatible with
 341 the structure storing a lossy or unrecognisable function of the signal, and a
 342 state that predicts is not thereby a state that means. The narrower claim is
 343 the one worth making: whatever internal states a physical structure holds,
 344 those that do not help predict its environment are paid for in heat, so the
 345 pressure toward states indexed to the environment is energetic rather than
 346 semantic. We call the phenomenon *structural resonance*, and there is no
 347 teleological drive to “survive” in it: predicting the environment is simply the
 348 cheaper of the two ways to hold a memory.

349 A separate gap lies between the stationary threshold theorems of Deriva-
 350 tion 10 and any statement about trajectories: the threshold $K_c = 2D$ is
 351 proved for a stationary density, not for the finite-oscillator system, and the
 352 dynamical mean-field limit — propagation of chaos for the Kuramoto sys-
 353 tem — is the missing link between them (Lacker, 2016; Sznitman, 1991). No
 354 theorem in this development closes it.

355 **7 Macroscopic scaling: a theorem and a com-** 356 **mitment**

357 Three steps of different kinds meet here and are easy to run together: con-
 358 vergence of a scalar energy, which is a theorem; introduction of a continuum
 359 coupling kernel, which is a modelling assumption; and identification of that
 360 kernel with the brain’s endogenous electromagnetic field, which is an empir-

361 ical claim about cortex. Neither later step inherits the licence of the first.
362 Kept apart, the empirical claim is the framework’s central commitment and
363 the place where it is most easily wrong — which is where a claim of this kind
364 belongs.

365 **The formal half.** `Phase2_MeshConvergence.lean` proves that the discrete
366 coupling energy converges to a continuous integral along regular refining
367 triangulations (`mesh_refinement_convergence`). Because the summand is
368 symmetric, the double sum is the midpoint rule over unordered edges, so the
369 discrete energy is a quadrature of the continuous one. What this does not
370 establish: it is a statement about an energy functional, not about dynamics
371 — no trajectory convergence and no renormalization-group flow. The con-
372 tinuous object, `ContinuousNeuralField`, is *posited* with a coupling kernel
373 K , not produced by the theorem.

374 **The empirical step.** The formal half supplies scalar energy convergence
375 and no kernel. The field sections therefore posit a symmetric continuum
376 kernel as a modelling input. Which physical variable in cortex realizes that
377 independently supplied kernel is a further empirical question, and we state
378 our answer as a commitment:

379 *The model identifies its posited continuum kernel with the endoge-*
380 *nous electromagnetic field of mammalian cortex.*

381 This is a claim about biology, not a consequence of anything above it.

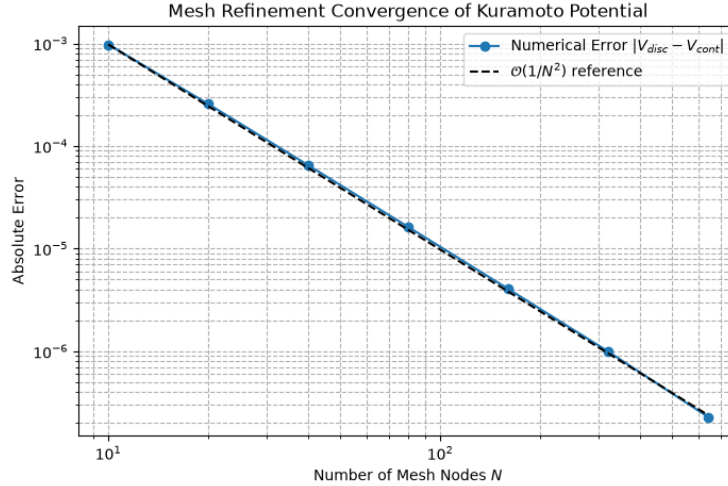


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of N units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field $\theta(x) = \sin x$, is compared against the continuous integral it is proved to converge to. The error falls as $\mathcal{O}(N^{-2})$, the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones. Parameters and scripts are in the supplement.

382 It is what makes the framework falsifiable and it is what we expect to be
 383 attacked.

384 **What supports it.** Neurons do not only signal through synapses: trans-
 385 membrane currents generate extracellular fields that shift membrane poten-
 386 tials of neighbouring cells, a mechanism known as ephaptic coupling. En-
 387 dogenous fields of 1–5 mV/mm entrain spike timing (Fröhlich & McCormick,

2010; Anastassiou & Koch, 2015), and stable electric fields carry information that survives representational drift (Pinotsis & Miller, 2022). The field is spatially continuous and superposes contributions from large populations — a medium in which spatially separated activity is combined without further mechanism. This is the integrative role the CEMI field theory assigns it (McFadden, 2002, 2020; Pockett, 2012), physically supported by astrocytic syncytia joined by gap junctions (Verkhratsky & Nedergaard, 2018).

What the evidence supports, and what it does not. The endogenous field’s effect on any individual neuron is modulatory — it biases when a cell fires; it does not decide whether it fires (Anastassiou & Koch, 2015). We do not claim it is the *primary* dissipative structure. The claim is that it is modulatory in effect and integrative in spatial reach: weak per synapse, and the only variable defined everywhere at once. What the framework needs is the reach, not the strength.

Why reach is the property that matters. A theory of consciousness needs an account of why many local processes make one experience, and most supply it by stipulation or by a measure. The account here is structural, and it is the framework’s strongest argument. The field is the only variable in the system that is *defined everywhere at once*. Unity is not asserted of it; unity is what a field is. That is what makes the sheaf-theoretic step of Section 8 available at all — local states glue to one global section because there is one object over which gluing can happen — so the answer to the combination

410 problem is a piece of geometry rather than a postulate. Compare “integra-
 411 tion” in Integrated Information Theory, which is a number, or “broadcast”
 412 in Global Workspace Theory, which is a metaphor about a workspace. It is
 413 also what the empirical identification is carrying: if the posited continuum
 414 coupling is realized by something that is *not* defined everywhere at once —
 415 by synaptic connectivity, say — then this argument is unavailable, and the
 416 framework loses the property that distinguishes it from its rivals rather than
 417 merely losing a mechanism.

418 **What the formal statement demands.** The identification is a hypothe-
 419 sis of the chain rather than a remark beside it. `IsEMFieldCoupling` (`Phase9_`
 420 `EMIdentification.lean`) is the predicate the arrow `E56` carries, and a model
 421 claiming the kernel is the endogenous field has to exhibit five things: a jointly
 422 continuous kernel; a normalised substrate, so that coupling strength is not
 423 a statement about how many sites the substrate has; a mean-field constant
 424 K that *is* that kernel averaged over both arguments rather than a num-
 425 ber set beside it; no single site contributing more than half of it, which is
 426 the modulatory condition above in formal dress; and a noise D that is the
 427 field’s own. It does not decide the biology — its witnesses are a unit inter-
 428 val and a three-site toy, and no object in the development denotes cortex
 429 — but it makes failure locatable: a kernel concentrated at one vertex fails
 430 the fourth condition, a substrate that couples nothing cannot discharge the
 431 third (`not_isEMFieldCoupling_of_zero_kernel`), and the threshold com-

432 mitment $K > K_c$ attaches to the same field the identification names (`em_`
433 `field_exhibits_phase_transition`).

434 **Its critics.** The longest-standing objection is that the fields are too weak
435 to matter, pressed from within the field-theoretic camp by Pockett herself
436 (Pockett, 2012, 2002). Vöröslakos and colleagues found that applied cur-
437 rent is shunted through scalp, delivering fields at or below 1 V/m in cortex
438 (Vöröslakos et al., 2018). That is a bound on *applied* rather than endogenous
439 fields, but it constrains how much causal work a field of that magnitude can
440 do, and is the reason we do not claim the field drives spiking.

441 **What would falsify it.** We state the failure conditions rather than leav-
442 ing them to a referee. **Epiphenomenality:** if the endogenous field can
443 be cancelled or shunted without altering transmembrane currents and nei-
444 ther behaviour nor report changes, the identification fails. **Magnitude gap:**
445 if the spatial integration required demands field effects an order of magni-
446 tude above the 1–5 mV/mm measured in cortex, the commitment fails on
447 arithmetic. **Timescale mismatch:** if recovery from sleep inertia tracks an
448 exponential rather than the delayed sigmoid the slow variable predicts, the
449 framework loses its own prediction. **Dissociation:** if large-scale field coher-
450 ence is preserved where unity is lost — or absent where it is retained — the
451 identification has no physical carrier.

Can a modulatory field cross the threshold? The objection that 1–5
 mV/mm is too weak to meet $K > K_c$ assumes that the per-neuron effect of
 the field is what enters the mean-field coupling constant. It enters through
 population arithmetic. A field of 3.0 mV/mm shifts spike timing by ~ 1.2
 ms (Anastassiou et al., 2011; Anastassiou & Koch, 2015); at 40 Hz, that is
 a per-neuron phase advance of $\Delta t \cdot f \approx 0.007$ per cycle. The LFP decay
 length $\lambda \sim 0.2$ mm (Lindén et al., 2011) encloses $N \approx 1676$ neurons at
 a cortical density of $5e + 04 \text{ mm}^{-3}$ (Herculano-Houzel, 2009). Integrating:
 $K \sim N \Delta t f \approx 80$. Against a Lorentzian frequency-spread floor $D \sim 1.5$
 rad/s (Strogatz & Mirollo, 1991; Buzsáki & Draguhn, 2004), the ratio $K/D \sim$
 53.6 clears the threshold $K_c/D = 2.0$ by a wide margin. The estimate is
 parametric in λ , which is empirically known and spans 0.1–0.3 mm (Lindén
 et al., 2011); across the full range and with field amplitudes at the measured
 bounds, the ratio stays above 30.2. Sweeping across canonical EEG bands
 with band-typical amplitudes (gamma: 0.5 mV/mm; beta: 1.0; alpha: 1.5;
 theta: 3.0; delta: 2.0), the same arithmetic yields K/D ranging from 8.9
 down to 1.8 — delta is the one band that sits below K_c , consistent with
 the loss of consciousness during slow-wave sleep. This is not a measurement,
 and it is the one commitment of the chain that is simplest to falsify on
 arithmetic alone: if the coupling parameters sit at the conservative end of
 each published range, K/D drops to 5.7 — still above threshold, but requiring
 the combination to survive independent measurement rather than being ruled
 out by the same arithmetic.

475 **The regime the theorems cover.** The dissipative account — well-posedness,
476 the Lyapunov identity, monotone descent, and that the motion stops along
477 every trajectory — holds with no assumption on the sign of the coupling.
478 *Where* it stops does not, and the claims about the destination all live on
479 the positive side. The memory capacity of a frustrated cortical network is
480 therefore not supported by any theorem here, and we do not assert it as part
481 of the chain. The natural repair — letting local states agree on overlaps up
482 to a phase and asking whether the offset cocycle is a coboundary — fails
483 because locked, twisted and splay configurations all assign a phase to every
484 patch, so their offsets are automatically a coboundary.

485 Primate LFP phase maps contain phase singularities with integer winding
486 number, and human cortical spiral centres classify cognitive tasks (Townsend
487 et al., 2015; Xu et al., 2023). `Phase5.PhaseLifts` formalizes the measured
488 loop: transitions from one global real lift telescope to zero, while a three-
489 patch loop with winding one admits no global lift. The supplement gives the
490 measurement protocol and preregistered test this supports — a falsifiable
491 content test outside the conditional chain.

492 **8 Global sections and the unity of conscious-** 493 **ness**

494 The system is now characterized by a continuous macroscopic electrodynamic
495 field generated by many coupled micro-components, forced to resonate with

496 the environment. From here on the coupling regime is the unfrustrated one,
 497 which is what every theorem below requires. The Kuramoto model (Ku-
 498 ramoto, 1984) guarantees that past a critical coupling threshold these dis-
 499 tributed local fields undergo a phase transition into unified synchronization
 500 (Bauer et al., 2022), and `Phase4.KuramotoDynamics.lean` confirms that the
 501 phase-locked state minimizes the Lyapunov potential — in the precise sense
 502 that the functional minimized is the coupling potential read in the frame ro-
 503 tating with the natural frequency, which `Phase4.RotatingFrame.lean` shows
 504 is the honest reduced description of a system of identical oscillators. With
 505 any natural frequency non-zero the full potential is unbounded below, so it
 506 has no minimum for phase-locking to attain; the reduction is exact only for
 507 identical frequencies, and a genuine spread leaves residual detunings whose
 508 threshold is nowhere in our development.

509 The phase transition supplies synchronized phases. A separate compat-
 510 ibility hypothesis says the corresponding local states agree on every over-
 511 lap; given that, the sheaf condition glues them into a singular *global section*
 512 (`Phase5.GlobalSection.lean`). The field acts as a single, indivisible topo-
 513 logical space, providing the *unity of consciousness*—a necessary condition
 514 for integrated experience. `ThermodynamicCover` demands that the cover’s
 515 phase field minimize the reduced Kuramoto potential, and `potential_min_`
 516 `iff_phase_locked` characterizes those minimizers exactly, in both directions.
 517 Every instance is therefore already phase-locked, and the class’s single con-
 518 dition on its local data — that patches at a common phase agree where

519 they overlap — makes the family compatible. `probability_glue_unique`,
 520 the sheaf property of the probability presheaf, which contains no physics at
 521 all, then glues that family into exactly one global section.

522 The shape of the class makes this an emergence theorem rather than
 523 a uniqueness theorem. Were the class to carry a global section as a field,
 524 together with a second field declaring every local section a restriction of it,
 525 the global object would exist before the sheaf condition was consulted. It
 526 carries neither: the global section is constructed, and being a restriction is a
 527 *theorem*. The witness makes this concrete: two patches carry mass profiles
 528 $(2, 1, 7)$ and $(5, 1, 3)$ across three sites, agreeing only at the shared site. The
 529 theorem returns profile $(2, 1, 3)$ — neither of the two, and neither restricts
 530 correctly to both patches. The global state carries information no single
 531 patch had.

532 Two hypotheses carry physical content. That synchronized patches agree
 533 on their overlaps is a modelling assumption, not a consequence of any dy-
 534 namics we formalize. The other, `thermodynamic_equilibrium`, asks that the
 535 cover already sit at the potential minimum. That one is discharged rather
 536 than assumed, and the route is worth stating because it determines the scope
 537 of the unity claim. `kuramoto_cauchy_problem` proves the Kuramoto initial
 538 value problem well posed on all of $\mathbb{R}$, so through every initial state there
 539 passes exactly one trajectory; along every trajectory of a zero-frequency sys-
 540 tem the dynamic potential converges and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is
 541 finite; and since the dissipation rate is Lipschitz in time, Barbalat’s lemma —

542 which Mathlib does not carry, and which is proved here — gives $\sum_i \dot{\theta}_i^2 \rightarrow 0$,
 543 so the motion stops along every trajectory. *Where* it stops is taken condition-
 544 ally, by a route that avoids the LaSalle invariance principle, which Mathlib
 545 also does not carry. The flow is a gradient flow, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an
 546 identity rather than an inequality; the minimizers of V are characterized ex-
 547 actly; and on the region where every pair of phases lies within a quarter turn
 548 a Polyak–Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, with W the
 549 excess above the minimum and $a > 0$ a lower bound for the coupling. The in-
 550 variant region is produced by the Lyapunov function rather than assumed: a
 551 single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so an excess below $a/2$ forces
 552 every pairwise cosine positive for all time. Descent plus the estimate inte-
 553 grates to $W(t) \leq W(0)e^{-ct}$, which makes each $|\dot{\theta}_i|$ integrable on $[0, \infty)$, and
 554 a trajectory with integrable velocity converges. The conclusion (`kuramoto_`
 555 `tendsto_global_minimum`) is that if the phases start within a quarter turn
 556 of each other and the initial excess is below $a/2$, the trajectory converges to
 557 a phase-locked configuration which is a *global* minimizer, with order param-
 558 eter $r^2 = 1$. `ThermodynamicCover.ofConvergentTrajectory` then takes a
 559 cover whose phase field is the limit of such a trajectory and discharges the
 560 equilibrium field *by that convergence*, with the cover’s coupling matrix the
 561 coupling matrix of the dynamics rather than a separately chosen one. On
 562 the resulting three-patch witness the starting configuration is provably not
 563 phase-locked, so the configuration unity requires is arrived at rather than
 564 posited.

565 **What the compatibility hypothesis says.** The remaining assumption
 566 is stated categorically — two patch-local sections have the same restriction
 567 to their overlap — and on a discrete substrate it has a pointwise reading,
 568 which is what makes it a physical claim rather than a bookkeeping con-
 569 dition. `restrict_eq_iff_density0n_eq0n` proves the two equivalent: the
 570 restrictions agree exactly when the two patches assign the same mass to ev-
 571 ery site they share. So the hypothesis says that patches whose phases agree
 572 modulo 2π carry the same local density where they overlap. It excludes
 573 something concrete: a two-patch cover of the three-site substrate, carrying
 574 one phase and two profiles that differ at the shared site, satisfies every other
 575 requirement — the patches cover the substrate, each section is a section over
 576 its own patch — and fails this one (`overlap_agreement_fails`). No dynam-
 577 ics in the development excludes such data; the assumption does. Note also
 578 what the hypothesis does *not* say. Patches may carry different content where
 579 they do not meet: the three-patch witness has three pairwise-different pro-
 580 files and satisfies the condition, because they differ only outside the overlaps.
 581 Agreement on overlaps is consistency, not sameness.

582 The two hypotheses cannot be dropped, and this is the honest scope of
 583 the result. Splay and twisted configurations are equilibria of the same flow,
 584 so “every trajectory reaches the unified state” is false as a general statement,
 585 and the initial-data condition is the standard arc condition under which
 586 the true statement holds. What is claimed is conditional — unity is reached
 587 from configurations already sufficiently coherent — and not that an arbitrary

588 initial state is carried into unity.

589 **Against IIT and GWT.** Our global section plays a structurally analogous
590 role to IIT’s Φ (Tononi & Koch, 2016) but is arrived at from the opposite end:
591 forced by a thermodynamic phase transition rather than postulated from phe-
592 nomenal axioms. The unity condition is checkable from field dynamics rather
593 than requiring the intractable computation of Φ , and exclusion falls out —
594 gluing either succeeds or it does not. Both are vulnerable to the objection
595 that the criterion could be satisfied by a system with no claim to experience
596 (Aaronson, 2014); our reply is that gluing is necessary but not sufficient, and
597 that the topological boundary and reflexive structure are additional required
598 conditions. Global Workspace Theory (Baars, 1997; Dehaene & Marti, 2014)
599 specifies a functional architecture without specifying what physical property
600 makes broadcast content conscious (Block, 1995). Our framework supplies
601 the missing substrate: the continuous field *is* the medium of integration, and
602 the $K_c = 2D$ threshold gives a physical condition for ignition. Where GWT
603 is agnostic about implementation, we make a falsifiable commitment — unity
604 requires a continuous physical medium — which predicts that a functionally
605 equivalent discrete broadcast architecture would not be conscious.

606 **Dynamic Core and predictive binding.** Two further accounts share
607 enough of our premises that distinguishing them clarifies what the sheaf-
608 theoretic contribution is. The Dynamic Core hypothesis (Edelman & Tononi,
609 2000; Tononi & Edelman, 1998) holds that a distributed neural ensemble

610 whose members sustain re-entrant interactions among themselves more strongly
 611 than with the rest of the brain constitutes a functional cluster whose activ-
 612 ity correlates with conscious content. That premise — synchrony over a
 613 distributed set — is one we share. Where we differ is in what makes a
 614 cluster a unity: the Dynamic Core is identified operationally (the cluster
 615 that passes a mutual-information test), while the present framework iden-
 616 tifies it topologically (the cluster whose phases glue to a single global sec-
 617 tion). The distinction matters because the sheaf condition makes unity a
 618 binary state — sections either glue or they do not — while the Dynamic
 619 Core remains a graded property depending on a threshold for the mutual-
 620 information test. Predictive-processing accounts of binding (Friston, 2005;
 621 Hohwy, 2013) treat unified perception as the consequence of a hierarchi-
 622 cal generative model whose top-down predictions resolve bottom-up sensory
 623 ambiguity. The present framework agrees that inference — in the thermo-
 624 dynamic sense of Still’s bound — is what drives the internal state toward
 625 the environment’s statistics, but it gives the binding mechanism a different
 626 structure: global sections are constructed from local compatibility condi-
 627 tions rather than from prediction error minimization, and the unified state is
 628 a fixed point of the field’s own dynamics rather than of a recognition model.
 629 Each alternative accounts for a different aspect of the phenomenon — the
 630 Dynamic Core for the distributed-synchrony observation, predictive binding
 631 for the inferential character of perception — and the present framework’s
 632 contribution is to provide a formal language in which the unity claim itself

633 can be stated precisely enough to be falsified.

634 9 Reflexive topology and the emergence of 635 the Self

636 While a global section provides unity, a Self requires reflexivity. The unified
637 field is a dissipative structure that must resonate with external perturbations
638 while managing its own internal phase transitions, and time evolution drives
639 it to minimize thermodynamic friction with its own state changes. This
640 mandates *auto-resonance*: the field must predict itself. The Self emerges as a
641 higher-order topological feature when the internal geometry folds to include a
642 low-dimensional avatar of its own computational and causal boundary within
643 its predictive model (`Phase6_ReflexiveTopology.lean`) (Levin, 2019).

644 **What carries the weight.** `reflexive_topology_implies_self` is the
645 Banach fixed-point theorem with the physical commitments in its hypothe-
646 ses: the space of global sections is assumed nonempty, metric and complete,
647 and the predictive map is assumed to contract. All four are discharged on
648 the three-site substrate, and the metric is constructed rather than posited.
649 The construction first pierces the sheafification: on a discrete substrate every
650 point has a smallest open neighbourhood, so the mass a germ carries at a
651 point is well defined, and `massEquivOn` proves that over any open the sections
652 of the probability sheaf *are* the finite measures on that open, read as densities

653 against counting measure. Mathlib supplies no inverse to this sheafification,
 654 so the correspondence had to be constructed. The metric on sections is then
 655 the uniform distance between densities, proved complete by exhibiting the
 656 limit section of a Cauchy sequence rather than by the degenerate route of
 657 showing Cauchy sequences eventually constant. The metric choice is load-
 658 bearing: under the 0/1 metric every contraction is constant (`contracting_`
 659 `implies_const`), and the Lévy–Prokhorov metric does not support this ar-
 660 gument on a discrete substrate, because the ε -thickening of a set is the set
 661 itself for $\varepsilon < 1$.

662 **The map is the avatar, or there is no map.** The predictive map is the
 663 composite of the avatar’s encoding and read-out, not free data alongside it.
 664 With the map free, Banach’s fixed-point theorem projects it out and never
 665 touches the avatar — replacing the avatar by a constant leaves the same Self.
 666 Three consequences follow from the composite structure. The Self lies in the
 667 range of the read-out (`self_mem_range_readout`), so it is reconstructed from
 668 a single section over the avatar region. Under the resonance condition $s =$
 669 $\text{readout}(s|_{\text{avatar}})$: the field is what its own localized self-encoding reconstructs.
 670 Blinding the avatar *moves* the Self: the blinded boundary’s map is constant,
 671 and the original Self is not a fixed point. If a family of resonant avatars *covers*
 672 the substrate then the global state is determined by their local readings alone,
 673 so unity is recovered from the avatars by the same sheaf condition.

674 **The contraction law is an independent modelling postulate.** The
675 rate $\exp(-(K - K_c)\tau/2)$ with $K_c = 2D$ is the linear relaxation rate of the
676 mean-field order parameter over an interval τ . The framework independently
677 assumes that the avatar-mediated self-prediction map is Lipschitz at this
678 same rate; no oscillator or field theorem derives that bound. Given the
679 postulate, the threshold results determine whether Banach can use it: above
680 threshold the rate is below one, so the assumed bound yields a unique fixed
681 point; at or below threshold the rate is at least one, and contraction at that
682 rate is false for every self-map of every metric space. This says only that
683 the Banach route is unavailable below threshold, not that a Self cannot exist
684 there. The postulate is witnessed at $K = 3$, $D = 1$, $\tau = 2 \ln 2$, but designing
685 a witness to realize the factor is not a derivation of self-model dynamics
686 from oscillator dynamics. The rate may be derivable from a thermodynamic
687 consistency requirement — if the map contracted any slower, the friction of
688 updating the avatar would exceed the entropy the external drive dissipates
689 — but that derivation is not given here and the rate remains a postulate of
690 the framework.

691 **The fixed point is the glued state.** The standalone Banach theorem
692 produces a fixed point somewhere in the space of global sections; by itself it
693 does not say that this is the section the sheaf condition glued. `extttChain.lean`
694 preserves that identity. `extttIsUnifiedBy T s` says that s restricts to every
695 local state of the cover T , and the endpoint `extttUnifiedSelf` says that this

696 same s is the unique fixed point of the reflexive map. The edge hypothesis
 697 must therefore supply the equality $\text{extttpredict } s = s$ as well as the Lips-
 698 chitz bound; Banach supplies uniqueness but not the equality between the
 699 thermodynamic and reflexive constructions. On the witness, the two-patch
 700 cover glues exactly to the state that is already the unique fixed point of
 701 the non-constant avatar map. This exposes, rather than removes, the phys-
 702 ical identification the composition requires. What is still not established is
 703 that any field dynamics produces this particular read-out or performs the
 704 restriction, and that a cortex has the coupling the witness uses — the pa-
 705 rameters are a choice of units, and nothing here derives a coupling constant
 706 from cortex. It is worth distinguishing what extttUnifiedSelf means in the
 707 formalization from what it does not mean. (i) Within Lean it is the unique
 708 global section that satisfies two equations — the unification equation (it re-
 709 stricts to every local state of the cover) and the fixed-point equation (the
 710 avatar-mediated map returns it). (ii) It is not the first-person experience it
 711 is stipulated to be: that is $n10$ and sits outside the chain. (iii) It is not any
 712 fixed point of any map on any metric space; the Banach theorem supplies
 713 a unique attractor, but the claim about experience is that its content is the
 714 structure of the unified state, not that it exists as a mathematical object.
 715 (iv) It is not a claim about a conscious self in the psychological sense — the
 716 object is a thermodynamic and topological feature, and its identification with
 717 a subject is a philosophical commitment of the framework, not a theorem.

718 10 Continuous stochastic thermodynamics and 719 the field

720 To move beyond finite graph analogies, we model the macroscopic electro-
721 dynamic field as a *continuous neural field*, extending the discrete Kuramoto
722 model to a continuous spatial coupling kernel $K(x, y)$ (`Phase8_ContinuousField.lean`).
723 Subject to thermodynamic noise D , the system undergoes non-equilibrium
724 stochastic thermodynamics, and rather than a static Landauer bound the
725 field dynamics minimize an entropy production functional

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx. \quad (2)$$

726 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
727 dergo gradient descent on this functional, and that descent is a theorem
728 rather than a modelling assertion. The drift is a fixed ω plus a linear operator
729 applied to K , so σ is a quadratic functional of an affine image of the kernel. A
730 squared norm is Fréchet differentiable on any real inner-product space, and an
731 affine map is its own derivative, so the composite differentiates by the chain
732 rule (`hasFDerivAt_quadratic_of_affine`). The drift map as a bounded op-
733 erator $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$ exists by Cauchy–Schwarz, and its adjoint exists
734 because L^2 is complete, giving $\nabla \sigma(K) = (2/D) A^*(AK + \omega)$. `structural_`
735 `resonance_decreases_sigmaContinuum` is the descent result with no finite-
736 ness hypothesis in it, run on Lebesgue measure restricted to $(0, 1)$, a substrate

737 that is neither finite nor atomic. One limitation stays explicit: the phase field
 738 is held fixed throughout, so this is plasticity of the coupling at frozen phases,
 739 not joint (θ, K) dynamics.

740 A second result concerns where descent is headed, and is unconditional
 741 on kernel class. For a *reciprocal* kernel $K(x, y) = K(y, x)$, the integrand
 742 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric, so Fubini identifies its double integral
 743 with its own negative and the total coupling contribution vanishes. The total
 744 drift is therefore $\int \omega$ whatever the phases: coupling moves drift between sites
 745 but neither creates nor destroys it. Evaluating at the phase-locked config-
 746 uration pins its value, and the mean-matching hypothesis follows for every
 747 competitor (`phase_locked_minimizes_entropy_of_symm`). Reciprocity is the
 748 physically intended condition, but is not a formality: a *one-way* coupling on
 749 two sites is strictly beaten, halving σ .

750 **The threshold.** The phase transition to the unified global section oc-
 751 curs when the mean-field spatial coupling exceeds the thermodynamic noise
 752 threshold $K_c = 2D$, linking the emergence of unity to physical temperature
 753 and noise. That predicate requires the substrate’s measure to be a prob-
 754 ability measure, and the hypothesis is load-bearing rather than decorative:
 755 on a substrate of total mass m an unnormalized double integral carries a
 756 factor m^2 , so the comparison would track the substrate’s size rather than
 757 its coupling, and a system a factor of two *below* threshold can clear the un-
 758 normalized bound. Under normalization, a constant kernel has mean-field

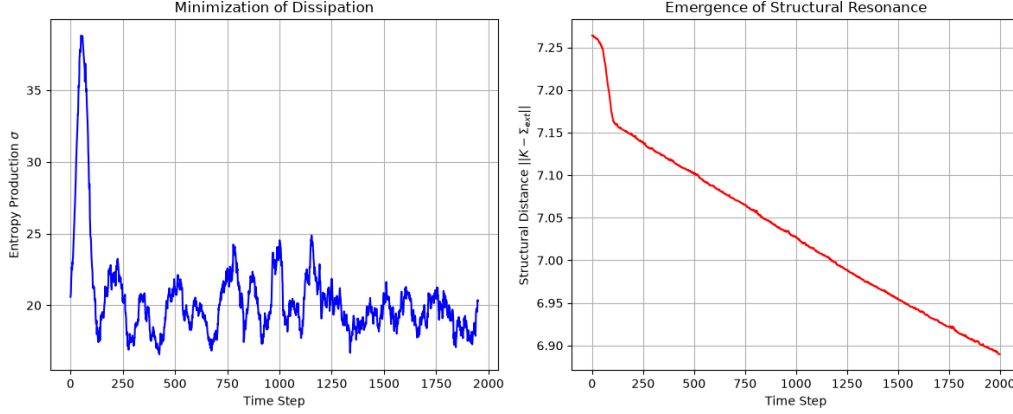


Figure 3: Coupling plasticity at frozen statistics: $N = 50$ units driven by perturbations drawn from a fixed environmental covariance Σ_{ext} , with the coupling matrix K descending the gradient of σ and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive σ , not a relaxation to zero. That plateau is the reason this section declines the inference from $\sigma \rightarrow \min$ to $D_{KL} \rightarrow 0$: the system visibly does not reach $\sigma = 0$. **Right:** the distance $\|K - \Sigma_{\text{ext}}\|$ decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on σ at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence. Parameters and scripts are in the supplement.

759 strength exactly K and the predicate reduces to $K > 2D$.

760 The threshold $K_c = 2D$ is not a bare stipulation. The stationary density
761 at order parameter r is von Mises $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency
762 requires $r = R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Formalized as a ratio of
763 integrals, integration by parts gives $\int \cos\theta e^{a\cos\theta} = a \int \sin^2\theta e^{a\cos\theta}$, so $R(a)$
764 is a times the mean of $\sin^2$ under the von Mises weight. The factor of two
765 comes from $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, proved by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ where

766 the integrand is a product of non-negative factors. At $K = K_c$ the map
 767 falls strictly below the diagonal for $r > 0$, so $r = 0$ is the only non-negative
 768 solution at threshold and the bifurcation happens strictly after K_c (`fixed_`
 769 `point_eq_zero_of_le_critical`).

770 Writing $R(a) = a \mathbb{E}_a[\sin^2 \theta]$, continuity of $\mathbb{E}_a[\sin^2 \theta]$ and $\mathbb{E}_0[\sin^2 \theta] = 1/2$
 771 give existence above $K_c = 2D$ by the intermediate value theorem (`supercritical_`
 772 `fixed_point_exists`). Uniqueness follows from $\mathbb{E}_a[\sin^2 \theta]$ being strictly de-
 773 creasing on $[0, \infty)$, proved without differentiation by folding both integrals
 774 onto $[0, \pi/2]$ and evaluating at a single crossing point. Together with conti-
 775 nuity at threshold this gives the qualitative bifurcation diagram.

776 Differentiation is avoidable for uniqueness and unavoidable for the *rate*,
 777 and the rate is what Section 12 needs, so `Phase8_CriticalExponent.lean`
 778 pays for it separately. Differentiating the cosine moments under the integral
 779 sign — dominated by the constant $e^{|a|+1}$ near the parameter — makes E twice
 780 continuously differentiable and evaluates $E(0) = 1/2$, $E'(0) = 0$ and $E''(0) =$
 781 $-1/8$ from elementary integrals of powers of $\cos$. Taylor's theorem gives
 782 $E(a) = \frac{1}{2} - a^2/16 + o(a^2)$, and substituting into the level-set equation gives
 783 the mean-field exponent: $r^2/(K - K_c) \rightarrow 1/D$ along any family of positive
 784 coherent solutions with $K \downarrow K_c$, so $\beta = 1/2$. Because those hypotheses hold
 785 only eventually along a filter, and an eventual clause is vacuously true on the
 786 empty one, the file discharges them on the branch itself rather than asserting
 787 that some family satisfies them.

788 `selfConsistency` is a function of r with a suggestive name. `vonMisesDensity`

789 is proved a genuine probability density on $[-\pi, \pi]$; `circularOrderParameter`
 790 is the average of $e^{i\theta}$ against a density; that average is proved exactly $R(a)$, real
 791 because the density is even; and `fixedPoint_iff_selfReproducing` states
 792 the equivalence the name promises: r solves $r = R(K, r)$ if and only if the
 793 von Mises density it induces has order parameter r . The discrete empirical
 794 average $\frac{1}{N} \sum e^{i\theta_j}$ (`order_parameter_complex`) and this continuum integral
 795 agree at the incoherent state (`incoherent_orderParameters_agree`): op-
 796 posite phases on two sites give $r = 0$, which is the order parameter of the
 797 uniform density. That is agreement at one configuration rather than an iden-
 798 tity, and it is worth exactly what it says — the two quantities are the same
 799 kind of object and are normalised alike — while the general link between
 800 them remains the mean-field limit, propagation of chaos. Read through this,
 801 the existence theorem says that above K_c there is a stationary density whose
 802 own order parameter is positive.

803 **What remains not established.** The stationary density is an input, not
 804 derived from dynamics. Of the coherent branch we prove existence, unique-
 805 ness, continuous emergence at threshold and strict growth in K , but not
 806 that it is dynamically selected. And the self-consistency equation is con-
 807 nected to a *density*, not a *trajectory*: relating the two is propagation of
 808 chaos for the finite Kuramoto system, a research programme rather than
 809 a lemma. That single gap is what separates our threshold predicate from a
 810 statement about dynamics. The derivation of the von Mises density from the

811 stochastic Kuramoto SDE follows a standard route that is not in question
 812 mathematically and is absent from Lean only because nobody has formalized
 813 the Fokker–Planck operator in Mathlib: write the SDE, write the Fokker–
 814 Planck equation for the N -particle density, factorise by exchange symmetry
 815 in the mean-field limit, and cite the known stationary solution of the result-
 816 ing McKean–Vlasov equation (Sakaguchi, 1988; Strogatz & Mirollo, 1991).
 817 The gap is formalization work, not a mathematical discovery, and nothing in
 818 the file depends on closing it: the threshold $K_c = 2D$ is proved conditional
 819 on the density, and a conditional statement is honest rather than speculative.
 820 Clinical evidence for this phase transition is observed during general anes-
 821 thesia: propofol disrupts high-frequency rhythm phase alignment, breaking
 822 structural resonance and forcing the integrated coupling below K_c (Bastos &
 823 Miller, 2021).

824 **11 Corollary: hardware thermodynamics and** 825 **topological rigidity**

826 The two results below are scoped to two distinct registers and must not
 827 be combined. `fieldCorrelation_sited_eq_zero` says that the continuum
 828 coupling energy functional registers exactly zero for any finitely-sited kernel
 829 — a statement about the functional, not about the architecture’s compu-
 830 tational power. `rigid_is_strictly_suboptimal` says that an architecture
 831 whose fixed wiring misses the current best-correlated pair is strictly subopti-

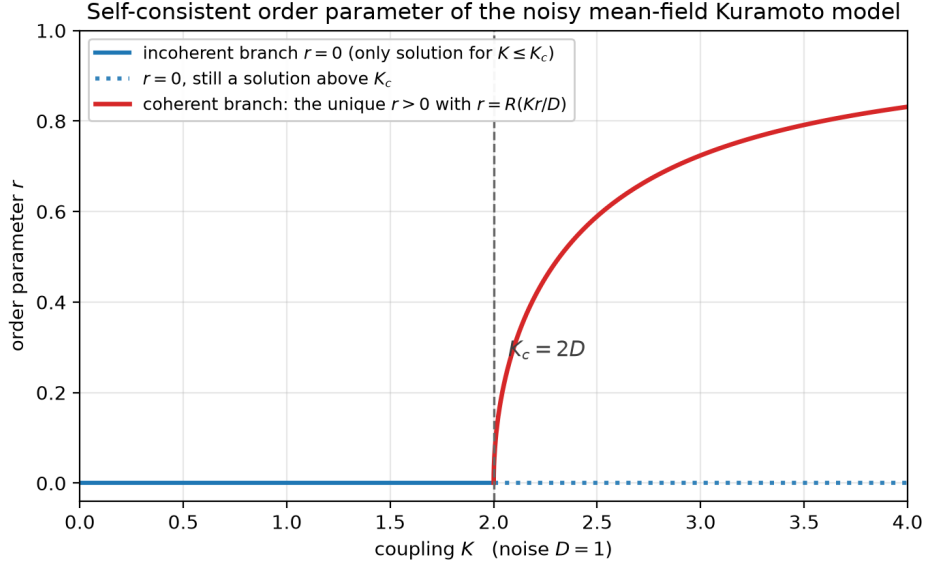


Figure 4: The self-consistency equation $r = R(Kr/D)$ of Section efsec:field, drawn at $D = 1$. Every feature of this picture is a proved statement about the equation. At or below $K_c = 2D$ the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one $r > 0$ joins it (`supercritical_fixed_point_existsUnique`), so the solution set is $\{0\}$ below and $\{0, r\}$ above; that branch tends to 0 as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in K (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving $E(a) = D/K$ for the concentration $a = Kr/D$ by bisection — legitimate precisely because E is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

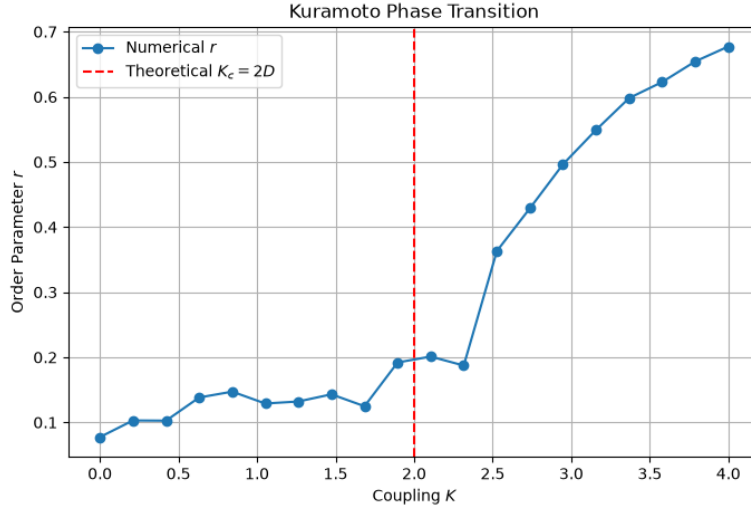


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here $N = 100$ oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale $D = 1$, and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is $K_c = 2D$ by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is $2D$ by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- N curve rises before the threshold and rounds off at it, as a finite sample must: $r \sim N^{-1/2}$ from incoherent phases alone.

mal at matched resource — a statement about a wiring condition, not about
all rigid architectures. Both qualifications are load-bearing and we restate
them in each derivation below. Both biological tissue and artificial silicon
hardware are physical dissipative structures. When a GPU computes, its
charge carriers undergo physical state transitions; by Landauer’s principle
(Landauer, 1961), this erasure exhausts heat into the environment. There is

838 a thermodynamic distinction in the degrees of freedom available to minimize
839 this dissipation: in standard von Neumann architectures the spatial topol-
840 ogy is rigidly fixed, and the coupling matrix A_{ij} is constrained by a discrete
841 layout that cannot be physically altered during operation.

842 The brain’s own architecture provides a suggestive constraint. The cere-
843 bellum contains approximately 80% of the brain’s neurons arranged in a dis-
844 crete, feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975),
845 and clinical observations of cerebellar agenesis report largely preserved con-
846 scious experience despite motor and cognitive deficits (Yu et al., 2015). This
847 dissociation has been read as evidence for the importance of recurrent archi-
848 tecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the level
849 of the anatomical observation while declining its Φ -based explanation. We
850 take the dissociation as evidence, not as confirmation: it is a single anatom-
851 ical case read against a theory, and it bears the weight such a case can bear.

852 Two formal results carry the corollary, and we state precisely what each
853 answers. The first objection is that a reallocation lemma can assume its
854 conclusion: a hypothesis such as “positive weight on a badly-correlated
855 pair while a better one exists” is most of what one wants to prove about
856 rigid hardware. It is not assumed here. `Phase7_Rigidity.lean` models
857 an architecture by the set S of site-pairs it has a *wire* for and requires
858 its couplings to vanish off S ; weights are tunable, wiring is not. Then
859 `totalCorrelation_le_of_realizable` bounds every realizable coupling by
860 $R \cdot M$, where R is the resource budget and M bounds phase correlation over

861 the wired pairs; the bound is attained; and `rigid_is_strictly_suboptimal`
 862 *derives* the suboptimality hypothesis whenever the substrate’s best-correlated
 863 pair is one the wiring misses, with `rigid_gap` making it quantitative. This is
 864 resource-matched and not question-begging, but note what it turns on: fixed
 865 wiring support, not continuity. The second objection is that a development
 866 in which both architectures are matrices on one finite index type does not dis-
 867 tinguish a continuum from a lattice at all. On a substrate whose measure has
 868 no atoms, `fieldCorrelation_sited_eq_zero` shows a kernel supported on
 869 finitely many sites contributes *exactly zero* coupling energy, whatever weights
 870 it carries and whatever the phase field does, while a patch of strength c on U
 871 achieves $c\mu(U)^2$; on the real line, a thousand units of coupling concentrated
 872 at the origin register zero against a unit patch that registers one. What that
 873 establishes is that the continuum coupling energy is blind to a finitely-sited
 874 architecture — a statement about which functional such an architecture can
 875 register in, not a demonstration that digital hardware computes worse.

876 A static phase field could in principle evade the wiring condition by being
 877 wired to precisely the best-correlated pair. But cortex is not static: the phase
 878 field evolves on sub-second timescales, and a rigid routing matrix cannot
 879 track it. Over any interval of non-zero length the field explores more than
 880 one correlation structure, so the wiring misses some optimal pair at almost
 881 every instant. The argument is informal — it does not need a formal ergodic
 882 theorem to state — but it closes a loophole that would otherwise leave the
 883 theorem silent about the dynamic case. The gap between the static theorem

884 and the dynamic conclusion is marked rather than blurred.

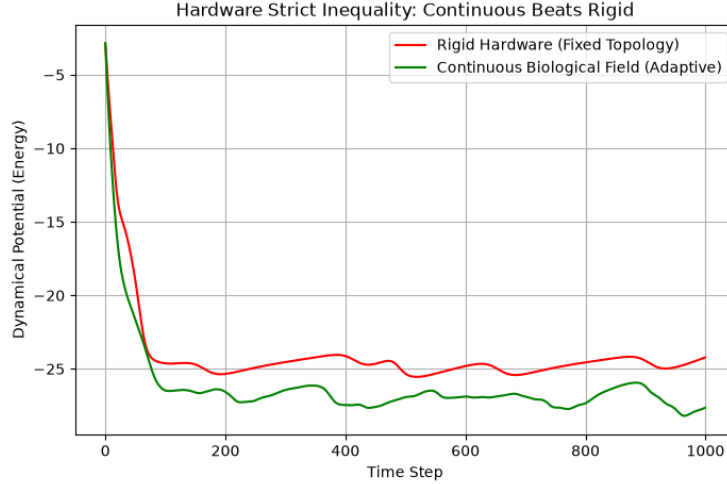


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same $N = 30$ oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that $\sum_{ij} A_{ij}$ is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, is the typical case rather than the general one. The 5% drift over 2000 steps at frozen phases reports a direction and not convergence. Parameters and scripts are in the supplement.

885 **What follows, and how far.** The step from either result to “von Neu-
 886 mann architectures cannot experience unified consciousness” remains an in-
 887 formal argument, which we mark rather than blur. On our reading, a GPU’s
 888 fixed routing matrix leaves it topologically fragmented, so the discrete layout

cannot supply the spatial coupling gluing into a global section requires, and rigid hardware cannot undergo the reflexive auto-resonance that generates a Self. This aligns with Searle’s biological naturalism: where the Chinese Room argument asserted that discrete symbol manipulation cannot generate understanding, our derivation supplies a candidate physical mechanism — topological rather than semantic (Searle, 1980). The obstacle to pure symbol manipulation is then physical rather than semantic: the substrate cannot hold a phase-locked field.

The constraint does not preclude artificial consciousness; it restricts its hardware. Analog field-based computing, neuromorphic architectures with continuous memristive plasticity, and thermodynamic computing systems (England, 2018; Ganesh & Parrondo, 2021) possess the spatial coupling required to physically morph their internal topology, which aligns with Hinton’s concept of “mortal computation” (Hinton, 2022): architectures where software is inextricably fused with the physics of the hardware. Biological consciousness uses macroscopic continuous electromagnetic fields together with a rich alphabet of molecular and transcriptomic cell states (Tasic et al., 2018), which allows the coupling to deform across a continuous parameter space.

A more precise thermodynamic bound can be stated, and it is the framework’s own arithmetic rather than a hand-waved distinction. A discretisation that approximates a continuum coupling kernel with N cells of diameter d must refresh each cell at the Nyquist rate of the field dynamics. Each refresh

912 that erases the previous state costs at least $k_B T \ln 2$ by Landauer's princi-
 913 ple. The product $N \times (\text{refresh rate}) \times k_B T \ln 2$ is a lower bound on the power
 914 density a silicon architecture must sustain to approximate the continuum the-
 915 orems. For cortical parameters — the number of effective degrees of freedom
 916 per cortical column, the field's characteristic frequency, and the metabolic
 917 cost of biological computation — the bound lies below the cortex's measured
 918 dissipation. For a GPU-scale discretisation with 10^9 transistors switching
 919 at GHz rates, the bound may exceed the device's thermal design power by
 920 orders of magnitude. The point is not the specific number, which depends
 921 on parameter choices that we do not fix here, but the existence of a princi-
 922 pled lower bound: a functionalist who claims that silicon can be conscious
 923 because it can simulate any continuum must first answer whether it can pay
 924 the thermodynamic cost the bound states. The bound is a rough estimate,
 925 not a theorem, and is marked as such.

926 **12 The sleep-inertia timescale mismatch**

927 Most of what this framework asserts is shared with some rival: that corti-
 928 cal synchrony matters, that dissipation constrains cognition, that unity is a
 929 physical rather than a computational property. One prediction is not shared,
 930 and it follows from the framework's specific commitment — that the medium
 931 carrying the unified state is a continuous field whose effective coupling de-
 932 pends on the physical geometry of the extracellular space, not on the rhythms

933 alone. Where that commitment differs from a purely electrical account, it
934 differs in an observable way, and we set out the difference precisely enough
935 to be wrong.

936 The daily loss of consciousness during sleep is modelled here as an active
937 physical disruption of the medium rather than a quiescence of the oscillators.
938 During sleep, astrocytic shrinkage expands the extracellular space, allowing
939 an influx of cerebrospinal fluid to flood the brain for metabolic waste clear-
940 ance (Xie & Nedergaard, 2013). This influx alters the spatial medium itself,
941 disrupting the continuous structural resonance and preventing the coherent
942 gluing that conscious unity requires on this account. The oscillators need
943 not stop for the field to fragment; the geometry they are coupled through is
944 what changes.

945 The prediction concerns sudden awakenings. When jolted awake, the lo-
946 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
947 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
948 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
949 cess fluid out of the interstitial pathways operates on a timescale of seconds
950 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
951 slower. During that lag the electrical oscillators are synchronizing across a
952 waterlogged topology, and the fragmented structural resonance is what is ex-
953 perienceed as grogginess. The unified global section re-establishes only when
954 astrocytic volume finishes matching the state of the electrical rhythms.

955 **Why this implies a delayed sigmoid.** Write $K(t)$ for the effective cou-
 956 pling, a monotone function of the reconstricting extracellular geometry, and
 957 $r(t)$ for the order parameter. The electrical variable is fast: at every instant
 958 the oscillators are at the configuration appropriate to the current K , so r
 959 tracks the equilibrium branch $r(K)$. The slow variable $K(t)$ carries the time
 960 course over 10^2 – 10^3 seconds. Section 10 establishes each piece as a theorem:
 961 at or below K_c only the incoherent solution exists; above it the coherent so-
 962 lution is unique, emerges *continuously* at K_c , and is strictly increasing in K .
 963 Composing monotone $K(t)$ with a branch that is zero below K_c , rises contin-
 964 uously from K_c , and saturates above yields three phases: a genuine *latency*
 965 while $K(t) < K_c$; a continuous rise as the threshold is crossed; and satura-
 966 tion. That is a delayed sigmoid, and the delay is not a fitting parameter —
 967 it is the time the slow variable takes to reach K_c .

968 **What the alternative predicts.** An account on which grogginess is the
 969 electrical realignment itself has no slow variable and no threshold. Its re-
 970 covery is exponential near the fixed point: monotone from $t = 0$, fastest at
 971 the start, with no flat foot. The discriminator is the presence or absence of
 972 an initial interval of no recovery, in data already collected (the high-density
 973 EEG awakening corpus of Stephan et al. (Stephan et al., 2025), with over
 974 1,000 awakenings from 20 subjects, is a suitable existing dataset for this test).

975 **The foot has a square-root cusp.** A delayed sigmoid is a shape that any
 976 two-timescale account with a slow saturating gate can produce, so the com-

977 parison above discriminates against less than it appears to. The bifurcation
 978 says more than that the coherent branch leaves zero continuously; it says
 979 at what rate. The second-order expansion $E(a) = \frac{1}{2} - a^2/16 + o(a^2)$, sub-
 980 stituted into the level-set form of the branch, gives the mean-field exponent
 981 in a path-independent form: along any family of positive coherent solutions
 982 with $K \downarrow K_c$ and $r \rightarrow 0$, $r^2/(K - K_c) \rightarrow 1/D$, that is $r \sim \sqrt{(K - K_c)/D}$,
 983 the pitchfork exponent $\beta = 1/2$. Composed with a coupling that crosses
 984 threshold at non-zero speed, so that $K(t) - K_c \approx \dot{K}(t_0)(t - t_0)$ just after
 985 the crossing time t_0 , the recovery leaves its foot as $r(t) \propto (t - t_0)^{1/2}$: a
 986 square-root cusp with *infinite initial slope*. An exponential has finite initial
 987 slope. So does a logistic. So does the composition of a smooth monotone
 988 gate with any branch that leaves zero linearly, which is what a generic sat-
 989 urating nonlinearity supplies. The cusp belongs to the bifurcation and not
 990 to the two-timescale structure, and it lives in the first instants after onset
 991 rather than in a fitted time constant — which makes it the sharpest available
 992 discriminator against a rival that reproduces the sigmoid by other means.

993 **A sharper form still, with no free parameters.** The stationary phase
 994 density at order parameter r is von Mises at concentration $a = Kr/D$, and
 995 the average of $e^{i\theta}$ under that density is exactly $R(a) = I_1(a)/I_0(a)$. Self-
 996 consistency says the density reproduces its own order parameter, so at every
 997 instant $r = I_1(a)/I_0(a)$, whatever $K(t)$ is doing. This yields three tests on a
 998 single recording. *Shape*: the phase distribution is von Mises, testable against

999 the histogram. *Collapse*: plotting $a(t)$ against $r(t)$, the whole recovery must
 1000 fall on $r = I_1(a)/I_0(a)$ — the *same* curve for every subject and session, since
 1001 it contains no subject-specific quantity. Individual differences move a point
 1002 along the curve, not off it. *Threshold*: $a(t)/r(t) = K(t)/D$ with no calibra-
 1003 tion, so the measured ratio must exceed 2 wherever coherence is non-zero
 1004 and approach 2 at onset. The two estimates must be made independently: a
 1005 from regressing the log-histogram’s slope on $\cos \theta$, r from the resultant length
 1006 $|\langle e^{i\theta} \rangle|$.

1007 **How to measure it.** The quantities the prediction constrains correspond
 1008 to specific neural observables. The order parameter r is the circular resul-
 1009 tant length of the instantaneous phase distribution across recording sites,
 1010 estimated from multisite LFP or source-reconstructed MEG/EEG with suffi-
 1011 cient cortical coverage. The concentration a is the slope of the log-density of
 1012 the same phase distribution regressed on $\cos \theta$, not the maximum-likelihood
 1013 estimate: fitting a to the resultant length would make the predicted col-
 1014 lapse a tautology. The slow variable $K(t)$ is the effective coupling strength,
 1015 whose dynamics the framework identifies with the extracellular volume frac-
 1016 tion — measurable via diffusion-weighted MRI during the sleep-inertia win-
 1017 dow, or proxied by astrocytic calcium imaging via fibre photometry. The
 1018 collapse prediction ties the two LFP-derived quantities (a and r) together on
 1019 a parameter-free curve, and the a/r threshold ties both to the volume proxy.
 1020 The measurement modality constrains the spatial coverage: at a minimum,

1021 electrodes or reconstructed sources spanning both hemispheres are needed
1022 to capture the distributed coherence the gluing theorem requires. A referee
1023 could preregister the analysis from the paragraph above and the protocol
1024 below, without emailing the author. The supplement reports the results of
1025 this analysis on human scalp EEG recorded during propofol sedation; the
1026 data are consistent with the collapse and validate the protocol, though over
1027 the narrow range of a that resting EEG occupies the curve is not yet distin-
1028 guishable from its own tangent, which is one more reason to run the protocol
1029 on a recovery that sweeps through threshold.

1030 A concrete dual-modal design pairs high-density EEG with rapid-acquisition
1031 diffusion-weighted MRI tracking the apparent diffusion coefficient (ADC)
1032 over the sleep-inertia window (10^2 – 10^3 s). The EEG stream supplies $r(t)$
1033 (circular resultant length) and $a(t)$ (log-density slope) at each time bin; the
1034 dwMRI stream supplies the ADC series as a proxy for $K(t)$. The parameter-
1035 free prediction is that the entire (a, r) recovery collapses onto $r = I_1(a)/I_0(a)$.
1036 The ADC-correlated prediction is that this collapse tracks the extracellular
1037 volume reconstruction through the threshold crossing $a/r = 2$. Both can
1038 be compared against an exponential null — monotone recovery from $t = 0$,
1039 fastest at onset, no flat foot — without fitting the collapse curve, because
1040 the curve contains no subject-specific parameter. At a minimum, the null
1041 comparison requires (i) a latency estimate (time from awakening to first non-
1042 zero coherence) consistent with zero in the exponential model and positive
1043 under the bifurcation model, and (ii) an onset exponent favouring $1/2$ over

1044 1 over the first 30–60 s of recovery, where the square-root cusp lives. The
1045 supplement gives the bootstrap procedure for both comparisons.

1046 **Falsifiability.** The parameter-free form tests the von Mises stationary den-
1047 sity and the mean-field closure jointly. It does not test $K_c = 2D$ indepen-
1048 dently of that ansatz: the crossing at $a/r = 2$ is a consequence of the same
1049 density, so a failure of the collapse and of the threshold are one failure. The
1050 prediction is falsified by: an exponential fit beating a delayed-sigmoid on
1051 early recovery; an onset preferring exponent 1 over 1/2; a latency that does
1052 not scale with extracellular volume; a phase distribution that is not von
1053 Mises; or an (a, r) trace departing from $I_1(a)/I_0(a)$. The first two falsify
1054 with no reference to the sleep-inertia setting at all.

1055 **Empirical check.** One falsification condition — departure of the (a, r)
1056 trace from $I_1(a)/I_0(a)$ — has been tested on human scalp EEG recorded
1057 during propofol sedation (OpenNeuro ds005620; 8 subjects, 62 electrodes,
1058 4–40 Hz; methods in the supplement). The order parameter r is the circular
1059 resultant length and the concentration a the log-density slope of the same
1060 distribution, estimated as set out above. So estimated, the data are consistent
1061 with the collapse: in a bipolar montage the mean residual is -0.0002 ± 0.0002
1062 with maximum $|r - I_1/I_0(a)| = 0.0013$ across the eight subjects, against
1063 an estimator noise floor of 0.0024. The check is weaker than those figures
1064 suggest, because resting EEG confines a to 0.302–0.542, an arc over which
1065 $I_1(a)/I_0(a)$ departs from a straight line by at most 0.0094; the data confirm

1066 that a and r stand in the required relation without yet distinguishing the
1067 curve from its tangent. This checks the phase-distribution claim, not the
1068 identification of the field with the coupling medium or the stipulation that
1069 the global section is experience.

1070 **No hysteresis.** The self-consistency equation has no fold: below threshold
1071 the only non-negative solution is $r = 0$; above it the non-negative solutions
1072 are exactly $\{0, r(K)\}$ with $r(K)$ strictly increasing from zero at K_c . There is
1073 no range of K carrying two distinct positive solutions, so the coherent branch
1074 emerges continuously and the descent into sleep and the recovery from it
1075 traverse the same $r(K)$ curve in both directions. This rules out any account in
1076 which a first-order transition separates the conscious and unconscious states
1077 — ignition- or avalanche-style models, for example — because a first-order
1078 transition would show a hysteresis loop between induction and emergence.
1079 The caveat is load-bearing: the development proves nothing about stability
1080 — $r = 0$ remains a solution above threshold — so the statement is about
1081 the solution set, not about dynamical selection. A system that stays on
1082 the incoherent branch despite the coherent branch being available would
1083 not show the predicted symmetry, and proving that it does not is an open
1084 stability question.

1085 **What this prediction assumes.**

13 Conclusion

Human-like cognitive architectures do not require biological mysticism or exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum theories of consciousness often invoke delicate subatomic phenomena to explain subjective unity, every component assembled here is classical: broken symmetries, finite phase space, Landauer erasure, and gradient descent on entropy production. What the formalization establishes is that the eight named hypotheses of the conditional composition are jointly consistent and that the framework’s remaining physical commitments are localized and named — not that classical fields are *sufficient* for consciousness or that the hypotheses follow from finite capacity. Two links carry that weight, and not in the same way. The unified state is proved to glue uniquely once a cover sits at the minimum of the coupling potential; the motion is proved to stop along every trajectory; and *where* it stops is proved conditionally, for initial data within a quarter turn and below an explicit energy threshold, with a cover built on that limit discharging the equilibrium hypothesis unity would otherwise assume. What is not proved is the unconditional statement, and it is false: splay and twisted configurations are equilibria of the same flow, so the basin condition is the scope of the claim rather than slack in it. That a physical cortex starts inside that basin is an assumption. The second link is untouched by any of this: the identification of a global section of the probability sheaf with the unity of experience is a stipulation of the framework,

1108 not a consequence of it.

1109 Philosophically, this framework bridges the historical gap between type
1110 identity theory (Smart, 1959) and computational functionalism (Putnam,
1111 1967). The subjective mental state is identical to the physical thermo-
1112 dynamic field, yet its qualitative nature is defined by the topological and
1113 functional dynamics of structural resonance rather than by the specific bio-
1114 chemistry of the brain. This constrains the *multiple realizability* of mind:
1115 consciousness is not exclusively bound to biological carbon, but neither is
1116 it substrate-neutral, because the substrate is what fixes which couplings are
1117 physically available. What is proved is that fixed wiring is strictly suboptimal
1118 at matched resource, and that a finitely-sited architecture registers exactly
1119 zero coupling energy in the continuum functional; the step from either to a
1120 claim about von Neumann architectures is an argument we make in prose
1121 and do not formalize. By locating the phase transition at a threshold with
1122 closed-form value $K_c = 2D$, this aligns with Seth’s “real problem” of con-
1123 sciousness (Seth, 2021): rather than bridging an unbridgeable metaphysical
1124 gap, the framework offers a topological architecture that can explain, predict,
1125 and physically perturb the structural properties of experience.

1126 **Code Availability**

1127 The complete Lean 4 formalization, encompassing the derivation of all theo-
1128 rems presented in this manuscript, as well as the accompanying Python sim-

1129 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1130 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1131 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1132 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/). Each of the five computed
1133 figures is produced by one script in `simulations/` and regenerates exactly;
1134 Figure 1 is drawn rather than computed.

1135 Contributions

1136 The conceptualization, manuscript drafting, and Lean 4 formalizations were
1137 heavily AI-assisted. The Google Antigravity CLI (Gemini 3.1, Claude 4.6
1138 Sonnet) served as a collaborative reasoning partner. DeepSeek V4 Flash
1139 and Claude Opus 5 (via OpenRouter/Nous Hermes Agent) assisted with
1140 code generation, proof debugging, and structural analysis. Claude Code
1141 carried out later formalization passes, figures, and manuscript restructuring.
1142 OpenAI Codex (GPT 5.6 Sol) contributed empirical analysis code and the
1143 (a, r) collapse pipeline. Responsibility for every claim rests with the author.

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